

Racial Disparities in Orange County Gang Enhancement Sustainments

Orange County District Attorney's Office Prosecutorial Data

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1 Introduction

The California gang database suffers from severe racial disparities and flawed metrics. Black and Hispanic males are disproportionately represented in these databases, while White males are significantly underrepresented. Though an individual is notified in writing before being entered/designated in a gang database in California, this practice is not a universal requirement. An individual may be in a gang database without their knowledge. Being included in a gang database can carry significant consequences, such as heightened police surveillance, barriers to employment and housing, and increased likelihood of sentence-enhancing gang allegations in criminal proceedings. California acknowledged the racial disparity, inconsistency, and unreliability in gang evidence and sentencing. This motivated the amending of Penal Code § 186.22 (gang enhancement statute) with Assembly Bill 333, effective January 1, 2022. This made the standards for applying gang enhancements more rigorous, to be used only when necessary and supported by sufficient evidence. California made a substantial effort to address the disparity; however, whether these reforms have translated into more equitable outcomes remains an open question. In particular, it is unclear whether the gang enhancement sustainments differ between Black and White defendants, or whether prosecutorial practices have changed across administrations. These unresolved questions motivate the present study.

1.1 Hypotheses

This paper examines racial disparities in gang enhancement sustainment outcomes between Black and White defendants. Using data from the Orange County District Attorney's Office,

2013-2019, 2021, and 2023-2024 were investigated. In addition to the ten-year analysis, an assessment of sustainment differences between former District Attorney Tony Rackauckas and current DA Todd Spitzer is included; however, the primary focus is on the racial disparities. Two hypotheses are tested to address these details specifically:

- **Hypothesis₁:** If gang enhancement allegations are guided using comparable evidentiary standards across racial groups, then conditional sustainment rates should not systematically differ between Black and White defendants. Systematic differences in sustainment rates may indicate differences in the evidentiary threshold used to pursue enhancements across racial groups.
- **Hypothesis₂:** If prosecutorial administration influences gang enhancement practices, then conditional sustainment rates should differ between the administrations of former District Attorney Tony Rackauckas and current District Attorney Todd Spitzer.

2 Background

Prior to examining the data, it is important to provide background on prosecutorial discretion and racial disparities in criminal justice.

First, sentence enhancements are used to add time to an individual's base sentence. California Penal Code § 186.22(b) is the gang enhancement offense code that can add an additional 2-10 years based on the felony or 15 years to life for specific severe crimes. Prosecutors are not legally required to pursue gang enhancements where eligible as they are the ones who decide if the evidence is strong enough to prove "without a reasonable doubt." The decision to pursue and maintain an enhancement involved substantial discretion. The discretionary authority has been identified as a potential source of unequal outcomes within the criminal justice system. California PC § 1385 is intended to prevent unfair prosecutions. The penal code allows judges to dismiss certain criminal charges or enhancements if doing so serves the interest in justice. Given the substantial discretion involved in alleging and sustaining gang enhancements, examining whether sustainment outcomes differ between Black and White defendant may provide insight into how prosecutorial decision-making operations in practice.

Second, Becker (1957) provides the theoretical framework that motivates the present study. In *Economics of Discrimination*, the author provides the classic economic theory of discrimination that distinguishes decisions drive by prejudice ("taste-based discrimination" and decision based on other incentives. Although Becker's analysis primarily focuses on

labor markets, the framework can be applied more broadly to situations in which decision-makers exercise discretion over similarly situated individuals. This is relevant to the present study because it provides a framework for examining whether defendants of different races receive different outcomes despite being subject to the same legal standards. This paper applies Becker’s theoretical insights to prosecutorial decision-making and gang-enhancement sustainments.

Third, Rehavi and Starr (2014) demonstrate racial disparities arise substantially during charging rather than solely during judicial sentencing. Gang enhancements are prosecutorial charging decisions. Examining the sustainments of this enhancement may reveal disparities originating earlier in the criminal process. Furthermore, Davis (1998) argues prosecutors possess considerable authority over charging decision and sentence-enhancing allegations. This discretion may contribute to racial disparities even when legal standards are race-neutral at face value. Davis does not evaluate gang-enhancement sustainments, which this study aims to analyze.

Based on these frameworks, this paper builds on the background by examining whether sustainment rates differ between Black and White defendants and whether the outcomes vary across DAs in Orange County.

3 Data Cleaning

3.1 Initial Cleaning: Raw Datasets

The Orange County prosecutorial data has inconsistent naming conventions for its various files throughout the years. Ensuring the cleaning process is accurate is critical to the analysis. For this paper and to account for the uncleanliness of the raw data, a R file was created to systematically clean, merge, and save the datasets from the repository. This section discloses the cleaning process part by part.

1. Download the necessary Excel Spreadsheets relevant for analysis per year

Of the files that are available on the OCPD website, the ones relating to charges filed, enhancements charged, priors alleged, sentences, and defendant information were applicable. Each file has various sheets, such as **Data Descriptions** that define the variables of the dataset and **Data** that stores the court information. The **Data** section was used for all the files.

After the files were read into the R environment, a series of joining the Excel Spreadsheets contributed to the final data frame of that year (**dat20XX**). The following explains the remaining steps necessary for cleaning the data at this stage:

2. Create new data frame `dat20XX` by filtering the `enhancements_charged20XX`'s column `Offense Code` to detect the Penal Code § 186.22 (California's Gang Enhancement), the unit of observation
3. Join `defendant_information20XX` by the `Defendant Number` to accurately match the case with the defendant
4. Join `charges_filed20XX` by the `Case ID Number`, `Defendant Number`, and `Count` to accurately match the criminal charge with the defendant information. This adds the primary charge to the enhancement, where both are specified, respectively
5. The `sentences20XX` Excel needs to be cleaned before joining the working data frame. `sentences20XX_clean` keeps the unique rows from the original data frame by the `Case ID Number`, `Defendant Number`, and `Count` to prevent duplicates. After this process, `sentences20XX_clean` is joined in the data frame (by `Case ID Number`, `Defendant Number`, and `Count`)
6. Similar to `sentences20XX`, `priors_alleged20XX` needs to be cleaned. `priors_summary20XX` is grouped by `Case ID Number` and `Defendant Number`, and a new column `has_prior` was created to flag if the defendant has a prior. The groups are dropped. After this process, `priors_summary20XX` is joined into the working data frame by `Case ID Number` and `Defendant Number`
7. Filter where Gender is not labeled "Unknown." This step finishes the cleaning process of `dat20XX`, where the dataset is exported into a file for later use

3.2 Data Preprocessing

After the initial cleaning of the years was completed, further steps needed to be taken before analysis. This section acts as a continuation of data cleaning with a priority of preprocessing the data for statistical evaluation.

8. Create `Year` column to specify the year of the original dataset
9. Create `Era` column to specify the era (Rackauckas or Spitzer) of the original dataset
10. Join all years' dataset into one data frame
11. Set the reference groups so `Race` is relative to `White`, `Year` is relative to 2013, and `Era` is relative to `Rackauckas`. Though R may default to these as the reference, explicitly declaring the comparison group prevents confusion

12. Create `sustained` column by flagging if the `Disposition_enhancement` column is true.
If the enhancement motion was accepted, then `sustained` = 1, 0 otherwise

4 Results

To gain a greater understanding of the topic as a whole, raw sustainment rates and logistic regression models were evaluated. The descriptive statistics provide an assessment of racial disparities in gang-enhancement outcomes. However, the raw rates do not account for the variation across years, prosecutorial administrations, or defendant characteristics. The logistic regression models are estimated to determine whether the observed disparities persist after controlling for those factors. Taken together, the descriptive and multivariate analyses provide a more complete assessment of racial differences of Black and White defendants in gang-enhancement sustainments.

4.1 Raw Gaps

The raw sustainment rate gaps were calculated throughout the ten-years. Figure 1 displays the raw sustainment rates between Black and White defendants for the years of study. Each bar provides the race by color (where White is red and Black is blue), the rate of sustains per race for the selected year, and the sample size of the race for the selected year.

From 2013 to 2017, Black defendants received more gang enhancement allegations than White defendants. However, White defendants generally had higher sustainment rates for these enhancements. This pattern suggests that Black defendants may have been charged with gang enhancements more frequently, despite those allegations being sustained less often.

2016 displays a reverse in the pattern, when Black defendants had a substantially higher sustainment rate than White defendants. This is the largest disparity of the study favoring Black with a difference of 39.9%. The year's sustainment rate for Black jumped from 7.5% to 64.5%. One possible explanation is in the California Supreme Court's decision in *People v. Sanchez* (2016). This case restricted prosecutors' ability to rely on "case-specific hearsay" through gang experts and required more direct evidence linking defendants to gang activity. Since the decision was issued in 2016, it may have altered how gang enhancement allegations were litigated and sustained during that time. This could potentially explain the shift, but additional analysis would need to be done.

Starting in 2018, White defendants are receiving more gang enhancement allegations than Black defendants. The sustainment rates for this year were the most similar of the ten-year analysis with a difference of 2.7%. Even with the similar rates, there were more

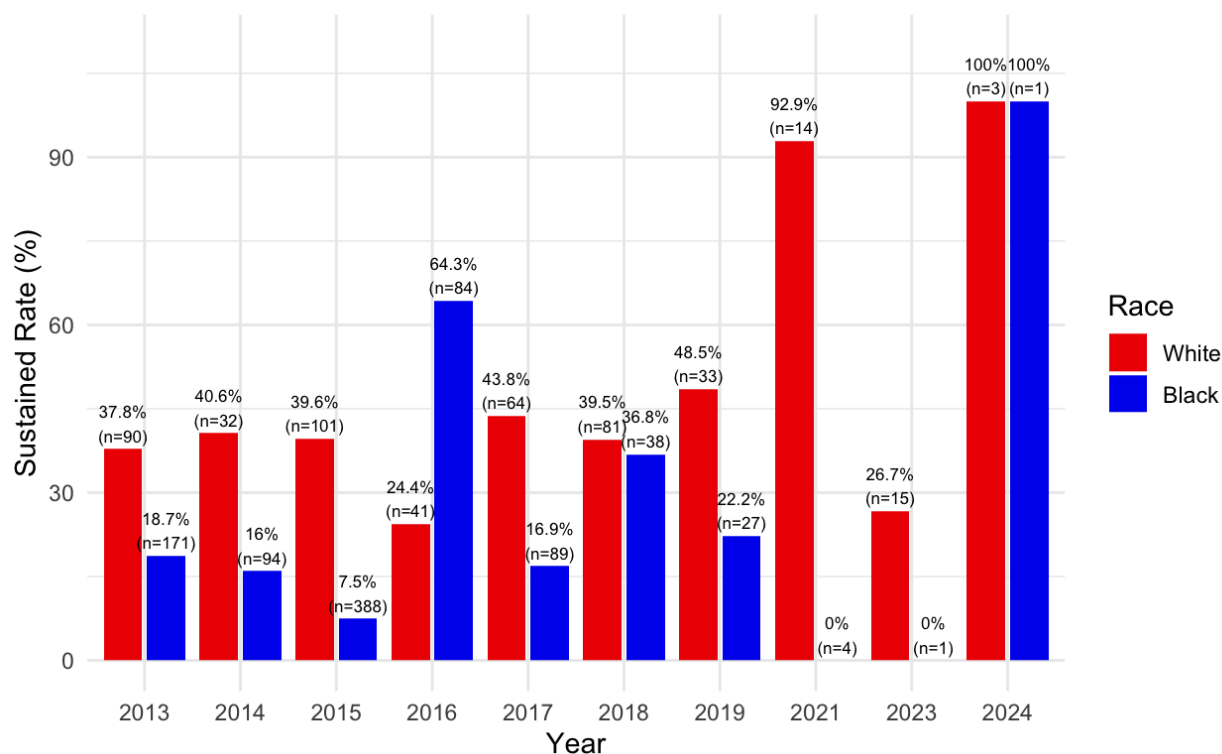


Figure 1: Raw Sustainment Rates Between Black and White Defendants

than twice as many White defendants ($n = 81$) as Black defendants ($n = 38$) who received gang-enhancement allegations. This may suggest that, based on raw sustainment rates, gang-enhancement allegations were upheld at similar rates across racial groups. However, the disparity in the number of allegations show the differences remained in who was being subjected to gang-enhancement charges.

2021 exhibits the largest racial gap favoring White defendants. There is a 92.9% difference in sustainment rates. However, this finding should be interpreted with caution. The sample sizes were substantially smaller than those observed in previous years. Only 14 White defendants and 4 Black defendants received gang-enhancement allegations during this year. Furthermore, 2023 and 2024 had fewer cases. A small sample size can make big impacts in calculated sustainment rates, making comparisons less reliable. These extreme disparities in the later years may reflect effects of limited sample size than meaningful change.

Briefly, during Rackauckas, White defendants had an average sustainment rate of 37.7% while Black defendants had a sustainment rate of 26.8%. During Spitzer, the former have a sustainment rate of 67% and the latter having a rate of 30.6%. The current DA's averages should be interpreted cautiously to account for the very small sample sizes.

4.2 Logistic Regression Models

Several logistic regression models were estimated in order to control for factors, such as race, year/era, and if the defendant has a prior.

4.2.1 Year

Table 1 displays the coefficient table predicting sustained outcomes. The formula with the most variables being controlled is the following:

$$\text{logit}(P(\text{sustained}_i = 1)) = \beta_0 + \beta_1 \text{Race}_i + \beta_2 \text{Year}_i + \beta_3 (\text{Race}_i \times \text{Year}_i) + \beta_4 \text{Prior}_i + \epsilon_i.$$

The coefficient of interest would be β_3 , the interaction effect between race and year. This measures whether racial disparities in sustain rates changed each year.

Across all four models that include a control for year, Black defendants have significantly lower odds of receiving a sustained outcome compared to White defendants. This relationship remains statistically significant even when controlling for priors alleged and race-year interaction effects. This suggests that racial disparities in sustainment outcomes persist throughout the ten-year study.

Several year effects are also significant in Models 1 and 2, 2015 is associated with lower odds of sustainment relative to the reference year (2013). 2016 and 2021 are associated with higher odds of sustainment. Yet, when it came to race-year interactions in Models 3 and 4, the main effect for 2016 is no longer statistically significant. This suggests that the rise in sustainment rates observed in 2016 may have been driven by race-specific differences.

The interaction terms provide further evidence of variation. For example, **RaceBlack** $\times$ 2016 interaction is positive and statistically significant. This suggests racial disparity temporarily reversed during this year, which is also seen in the raw rate analysis. In addition, **RaceBlack** $\times$ 2018 is also positive and statistically significant. The rest of the years were not statistically significant.

Finally, **has_prior** is positive and statistically significant in Models 2 and 4. Defendants with a prior allegation have greater odds of receiving a sustained outcome, even after accounting for race and year.

The results suggest that both race and temporal factors influence sustainment outcomes.

As a robustness check, **gender** was added to the logistic regression specification that did not include the interaction effects. The coefficient for **RaceBlack** remained negative and statistically significant ($\beta = -0.850, p < .001$). This indicates that the observed racial disparity is not explained by differences in gender across defendants. The analysis has remained unchanged after controlling for gender.

For another robustness check, 2016 was removed the data. The motivation behind this

Table 1: Logistic Regression Models Predicting Sustained Outcomes

	(1)	(2)	(3)	(4)
Intercept	-0.504** (0.164)	-0.872*** (0.196)	-0.499* (0.217)	-0.896*** (0.241)
RaceBlack	-0.961*** (0.139)	-0.962*** (0.140)	-0.970*** (0.293)	-1.078*** (0.296)
Year2014	-0.079 (0.263)	-0.031 (0.263)	0.120 (0.421)	0.281 (0.427)
Year2015	-0.597** (0.197)	-0.505* (0.199)	0.077 (0.298)	0.114 (0.301)
Year2016	1.206*** (0.235)	1.275*** (0.237)	-0.632 (0.424)	-0.672 (0.427)
Year2017	0.075 (0.235)	0.200 (0.240)	0.248 (0.333)	0.239 (0.336)
Year2018	0.321 (0.245)	0.401 (0.248)	0.073 (0.315)	0.069 (0.318)
Year2019	0.357 (0.312)	0.535 (0.315)	0.438 (0.411)	0.707 (0.420)
Year2021	1.705** (0.557)	2.017*** (0.564)	3.064** (1.060)	3.426** (1.065)
Year2023	-0.549 (0.601)	-0.449 (0.606)	-0.513 (0.623)	-0.462 (0.629)
Year2024	15.359 (430.540)	15.564 (438.335)	16.065 (840.274)	16.462 (840.274)
has_prior		0.494*** (0.140)		0.608*** (0.150)
RaceBlack $\times$ Year2014			-0.312 (0.543)	-0.450 (0.548)
RaceBlack $\times$ Year2015			-1.124** (0.405)	-0.995* (0.409)
RaceBlack $\times$ Year2016			2.689*** (0.519)	2.898*** (0.527)
RaceBlack $\times$ Year2017			-0.375 (0.479)	-0.049 (0.489)
RaceBlack $\times$ Year2018			0.857 (0.500)	1.131* (0.510)
RaceBlack $\times$ Year2019			-0.222 (0.649)	-0.393 (0.655)
RaceBlack $\times$ Year2021			-17.161 (727.700)	-17.193 (721.960)
RaceBlack $\times$ Year2023			-13.585 (1455.398)	-13.130 (1455.398)
RaceBlack $\times$ Year2024			0.970 (1680.548)	0.470 (1680.548)

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, . $p < 0.10$.

decision is 2016 being an outlier year where Black defendants had higher sustainment rates than White defendants. The coefficient on **RaceBlack** remained negative and highly statistically significant ($\beta = -1.360, p < .001$). The coefficient increased relative to the full-sample specification, which indicates that the observed racial disparity is not driven by 2016 results. If anything, excluding 2016 strengthens the estimated disparity between Black and White defendants.

The last robustness check filtered small sample sizes (2021, 2023, and 2024). The motivation behind this decision is reducing concerns about sample size bias. The coefficient on **RaceBlack** remained negative and statistically significant ($\beta = -0.916, p < .001$). This finding indicates that the observed racial disparity is not driven by the small sample sizes in later years of the dataset.

4.2.2 DA Era

Table 2 displays the coefficient table predicting sustained outcomes. The formula with the most variables being controlled is the following:

$$\text{logit}(P(\text{sustained}_i = 1)) = \beta_0 + \beta_1 \text{Race}_i + \beta_2 \text{Era}_i + \beta_3 (\text{Race}_i \times \text{Era}_i) + \beta_4 \text{Prior}_i + \epsilon_i.$$

The coefficient of interest would be β_3 , the interaction effect between race and DA era. This measures whether racial disparities in sustain rates changed after Spitzer took office.

The models that incorporate era indicators show that race remains a strong and statistically significant predictor of sustainment outcomes. Black defendants have significantly lower odds of receiving a sustained outcome compared to White defendants. This effect is robust.

EraSpitzer is positive and statistically significant across all models. This indicates that sustainment odds were higher during this period relative to the Rackauckas years. This suggests a shift in sustainment outcomes over time, holding race constant. Additionally, **has_prior** is consistently positive and highly significant, showing that defendants with prior involvement are more likely to have their gang enhancement allegation sustained.

While looking at the interaction, **RaceBlack** $\times$ **EraSpitzer** is negative in both interaction models. However, they are not statistically significant. This implies that though there is a weaker sustainment advantage during Spitzer for Black defendants relative to White defendants, the evidence is not strong enough to conclude racial disparities in these years.

Overall, the results indicate that the racial disparities in sustainment outcomes persist even when accounting for DA eras. The Spitzer era is associated with higher overall sustainment rates; however, this increase does not significantly differ across racial groups when interaction effects are included.

As a robustness check, **gender** was added to the logistic regression specification that

Table 2: Logistic Regression Models with Era Indicators

	(5)	(6)	(7)	(8)
Intercept	-0.452*** (0.099)	-0.735*** (0.129)	-0.473*** (0.102)	-0.778*** (0.133)
RaceBlack	-1.054*** (0.129)	-1.057*** (0.130)	-1.016*** (0.134)	-1.004*** (0.135)
EraSpitzer	0.538* (0.223)	0.660** (0.225)	0.689* (0.269)	0.877** (0.276)
has_prior		0.468*** (0.133)		0.489*** (0.134)
RaceBlack $\times$ EraSpitzer			-0.512 (0.512)	-0.705 (0.517)

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

did not include the interaction effects. The coefficient for **RaceBlack** remained negative and statistically significant ($\beta = -0.878, p < .001$). This indicates that the observed racial disparity is not explained by differences in gender across defendants. The analysis has remained unchanged after controlling for gender

5 Discussion

The intention of this paper was to examine whether gang-enhancement sustainment outcomes differ between Black and White defendants. An additional analysis studied whether those outcomes vary across prosecutorial administrations in Orange County.

The results provide evidence that sustainment outcomes were not distributed equally across racial groups. Both the descriptive statistics and logistic regression models indicate that Black defendants generally experience lower odds of gang-enhancement sustainment than White defendants, even after keeping year effects and prior allegations constant. These findings are consistent with Hypothesis₁. This suggests that gang-enhancement outcomes differed systematically by race during the period examined.

The race coefficient across multiple model specifications was critical. If gang enhancements were pursued and sustained under comparable evidentiary standards regardless of race, then sustainment rates would be expected to be similar after controlling for observable factors. Instead, Black defendants consistently exhibited lower odds of sustainment relative to White defendants. One interpretation is the evidentiary threshold for alleging gang enhancements may have not been uniformly applied across racial groups. Since sustainment represents the successful application of enhancement, lower sustainment rates for

Black defendants may indicate that enhancements were pursued in circumstances where the supporting evidence was relatively weaker.

To connect this paper back to the literature, Rehavi and Starr (2014) argue racial disparities frequently emerge during charging decisions instead of solely at sentencing. Gang enhancements are an example of prosecutorial charging decisions, and the results from this analysis suggest that disparities may persist in the initial decision to allege an enhancement and in the allegations sustaining. Additionally, Davis (1998) emphasizes that prosecutors possess substantial authority in determining which charges and sentence enhancements are pursued. The racial differences suggested in the study are consistent with the view that discretionary charging practices can contribute to unequal outcomes, even when the governing legal standards are race-neutral.

The year-specific results further exhibit that sustainment outcomes are not constant over time. Significant interaction effects in 2015, 2016, and 2018 suggest racial disparities varied across years, specifically a reversal in 2016. In this year, Black defendants experienced substantially higher sustainment rates than White defendants. This may have been connected to *People v. Sanchez*, but more research would need to be done.

In regard to Hypothesis₂, the results are quite limited. Sustainment rates were generally higher during the Spitzer era than during Rackauckas where the era coefficient was statistically significant across several models. However, the interaction between race and DA era was not statistically significant. Changes in prosecutorial leadership alone does not appear sufficient to explain the observed racial differences in sustainment outcomes.

These findings should be interpreted carefully. While the analysis identifies disparities in outcomes, it does not determine the mechanisms that are producing the disparities. There may be other factors, such as gang evidence, defendant criminal histories other than priors, or case complexity, that may influence sustainment outcomes that are not covered in the present analysis. The results should not be interpreted as direct evidence of intentional discrimination. This merely demonstrates the racial disparities existing in gang-enhancement sustainments persisting after controlling for observable factors. Reverting back to Becker, the findings are consistent with unequal treatment in discretionary decision-making.

Altogether, the results expose important questions about equity in gang-enhancement allegations. The findings suggests that disparities in sustainment outcomes existed throughout the ten-years examined but were not fully explained by changes in district attorney.

6 Limitations

Several limitations should be considered when interpreting the results of this paper. First, the years in the dataset exclude 2020 and 2022. This was a deliberate decision as there were unregulated reports in those two years. 2020 did not include **Offense Code** in its enhancements Excel Spreadsheet, which is the essential unit of observation. 2022 had an inaccurate **Defendant Number** column that did not match the rest of the Excel sheets. These reasons made it unusable for the present paper. Second, the years for comparing Rackauckas and Spitzer are uneven. The consequence of not including 2020 and 2022 is the former DA having two more years of analysis than the current DA. Lastly, the sample sizes began to exponentially decrease throughout the years, limiting the ability to generalize findings to a broader population. The decreased sample size may overestimate the true effect in the population.

7 Future Directions

Future work could address the limitations discussed and expand the findings of this paper. One such improvement would be to even the years for the Rackauckas and Spitzer investigation, or to include all the years in the original repository. The Orange County Prosecutorial Data includes years from 2000 to 2024, but only a decade was covered in this paper. Furthermore, a supplementary investigation on hate crime enhancements (PC § 422.75) may be worth comparing. White supremacist groups do not fall under the definition of 'gang' under the current interpretation of PC § 186.22. Defendants who are White supremacists often have the hate crime enhancement applied instead of the gang enhancement, where the latter enhancement can add more years than the former. Comparing the two would reveal impacts across racial lines, where identical/similar actions receive different penalties depending on the defendant's identity (gang or supremacist group).

8 Supplementary Materials

Project repository and source code are available at <https://github.com/jedsdino/ocpd-gang-enhancement-disparities>. The GitHub repository includes the Rmd files detailing the cleaning process and analysis used for this paper.

9 Conclusion

This paper analyzed whether racial disparities in gang enhancement sustainments differ between Black and White defendants. Using data from the Orange County District Attorney’s Office for 2013-2019, 2021, and 2023-2024, raw sustainment rate gaps and logistic regression models were evaluated. A thorough data cleaning process was completed to filter cases which alleged gang enhancements (PC § 186.22). Both descriptive statistics and regression models exhibit differences in sustainment rates throughout the years between the races. $\text{RaceBlack} \times \text{Year2015}$ is the only interaction term that is negative and statistically significant. The interaction terms for Black defendants and 2016/2018 are positive and statistically significant, indicating the racial disparity being temporarily reversed. As an extension to the study, the sustainment differences between former District Attorney Tony Rackauckas and current DA Todd Spitzer was evaluated. Racial disparities do persist in sustainment outcomes when accounting for DA eras. Yet, the Spitzer era is not statistically significant when interaction effects are included. Overall, the findings provide evidence consistent with Hypothesis 1 since sustainment outcomes differed systematically between Black and White defendants. The evidence for Hypothesis 2 is more limited. Changes in DA administration does not fully explain the observed racial disparities in gang-enhancement sustainments.

10 References

- Becker, G. S. (1995). *The economics of discrimination* (2. ed). Univ. of Chicago Press.
- Davis, A. (1998). *Prosecution and race: The power and privilege of discretion*. Fordham Law Review, 67(1), 13–67.
- People v. Sanchez, 63 Cal. 4th 665 (Cal. 2016).
- Rehavi, M. M., & Starr, S. B. (2014). Racial disparity in federal criminal sentences. *Journal of Political Economy*, 122(6), 1320–1354. <https://doi.org/10.1086/677255>